



Shahid Beheshti University

Department of Computer Science
Advanced Data Mining / Deep Learning

Mini Project 4

ViT-1: Vision Transformer Fine-Tuning

PlantVillage Classification with a ResNet-50 Baseline

Project Information

Architecture family:	Vision Transformer
Dataset:	PlantVillage
Project value:	30 points
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Student:	Alireza Jahanbakhsh
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Notebook Link:	Kaggle Notebook
Project Link:	GitHub Repository

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Data Mining Course

1 Problem and Data

This project classified PlantVillage crop-leaf images into 38 plant-health classes [3]. The executed notebook located a PlantVillage image-folder root on Kaggle and reported 43,444 images across the 38 class folders.

For a practical but controlled Kaggle experiment, the notebook did not train on the full original class distribution. Instead, it sampled at most 160 images per class using class-dependent seeded shuffling. This produced a capped subset of 6,041 images and reduced the original class-frequency imbalance. The subset was then split once with stratification into 4,228 training images, 906 validation images, and 907 test images. The same split was reused by every main model, and the test partition was left untouched until final evaluation.

All architectures received 224×224 RGB images. Training preprocessing consisted of resizing, random horizontal flipping, conversion to tensor, and ImageNet normalization with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). Validation and test preprocessing used deterministic resizing, tensor conversion, and the same normalization.

2 Method

2.1 ResNet-50 CNN Baseline

The CNN baseline was a pretrained ResNet-50 [2]. Its final fully connected layer was replaced by a 38-class classifier, and the entire network was fine-tuned in the main experiment. After classifier replacement, the model had 23,585,894 trainable parameters.

2.2 Vision Transformer

The Transformer model was a pretrained ViT-B/16 loaded through `timm` with model identifier `vit_base_patch16_224` [1]. Its classification head was configured for the 38 PlantVillage classes.

Three fine-tuning strategies were evaluated:

- **Head only:** all backbone parameters were frozen and only the classifier head was trained.
- **Last two blocks:** all parameters were first frozen, then the final two Transformer blocks, final normalization layer, and classifier head were unfrozen.
- **Full:** all ViT parameters were trainable.

The main models used cross-entropy loss, AdamW, batch size 16, learning rate 10^{-4} , weight decay 10^{-4} , and a two-epoch training budget. The best checkpoint within each run was selected by validation Macro-F1. The main comparison matched the data split, image preprocessing, batch size, number of epochs, optimizer family, and validation-selection rule. It did *not* equalize FLOPs or the number of trainable parameters.

3 Original Contribution

3.1 Hypothesis and Layer-Unfreezing Study

The original contribution was a controlled ViT layer-unfreezing experiment. The hypothesis stated in the notebook was:

“Partially unfreezing the final two ViT encoder blocks will improve validation macro-F1 over head-only training and remain within one percentage point of full fine-tuning, while updating substantially fewer parameters.”

The notebook defined the hypothesis as supported only if the last-two-block strategy (1) outperformed head-only training in validation Macro-F1, (2) remained within one percentage point of full fine-tuning, and (3) used fewer trainable parameters than full fine-tuning.

Table 1: ViT layer-unfreezing ablation on the validation split.

Strategy	Val. accuracy	Val. Macro-F1	Trainable params	Seconds
Head only	0.8775	0.8724	29,222	147.4528
Last two blocks	0.9647	0.9649	14,206,502	182.9765
Full fine-tuning	0.9272	0.9290	85,827,878	367.1911

The last-two-block strategy achieved the highest validation Macro-F1. It improved substantially over head-only training, exceeded the full-fine-tuning result under the same compact epoch budget, and updated far fewer parameters than full fine-tuning. Therefore, the notebook marked the hypothesis as **SUPPORTED** and selected last2 for final test evaluation.

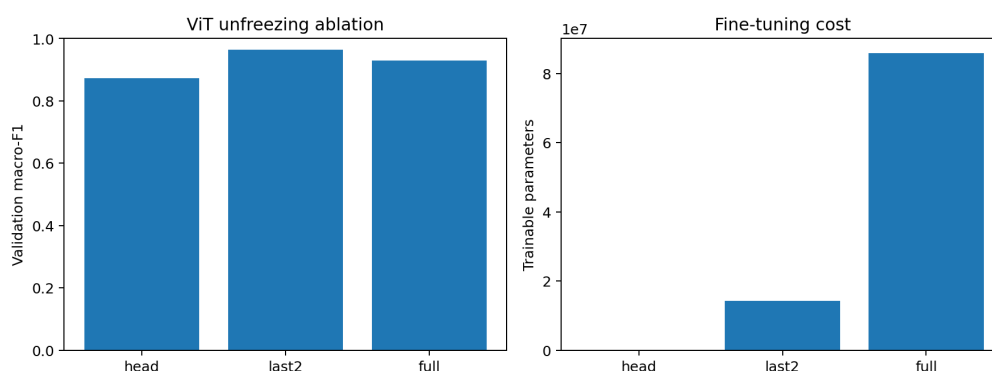


Figure 1: Validation Macro-F1 and trainable-parameter cost for the three ViT fine-tuning strategies.

4 Experiments

The experimental sequence followed the project requirements:

- train and validate the fully fine-tuned ResNet-50 baseline;

- compare the three ViT layer-unfreezing strategies and select the strategy using validation Macro-F1;
- evaluate the selected ViT and ResNet-50 on the untouched test split;
- perform paired statistical analysis because both models were evaluated on the same test images;
- analyze class-level F1 differences and representative prediction disagreements;
- compare ViT final-block CLS-to-patch attention with ResNet Grad-CAM on the same test images;
- run the handbook stretch experiment on data efficiency;
- run a controlled late-self-attention CNN experiment to answer the first required Discussion & Observation question empirically.

5 Results and Analysis

5.1 Final Matched-Protocol ViT versus CNN Comparison

The validation-selected ViT-B/16 (last two blocks) and the fully fine-tuned ResNet-50 were evaluated on the same 907 held-out test images.

Table 2: Final test comparison on the held-out split.

Model	Accuracy	Macro-F1	Trainable params	Seconds
ViT-B/16 (last2)	0.9713	0.9713	14,206,502	182.9765
ResNet-50	0.9691	0.9691	23,585,894	106.5869

The point estimate slightly favored ViT, but the paired analysis does not justify a strong superiority claim. ViT was correct while ResNet was wrong on 23 test images; ResNet was correct while ViT was wrong on 21. The two-sided exact McNemar-style test used in the notebook returned $p = 0.8804$. The observed Macro-F1 difference, ViT minus ResNet, was 0.00213, and the paired 2,000-resample bootstrap 95% confidence interval was $[-0.01242, 0.01619]$. Because the interval includes zero and the paired p-value is large, the final test results are consistent with essentially comparable performance under this protocol.

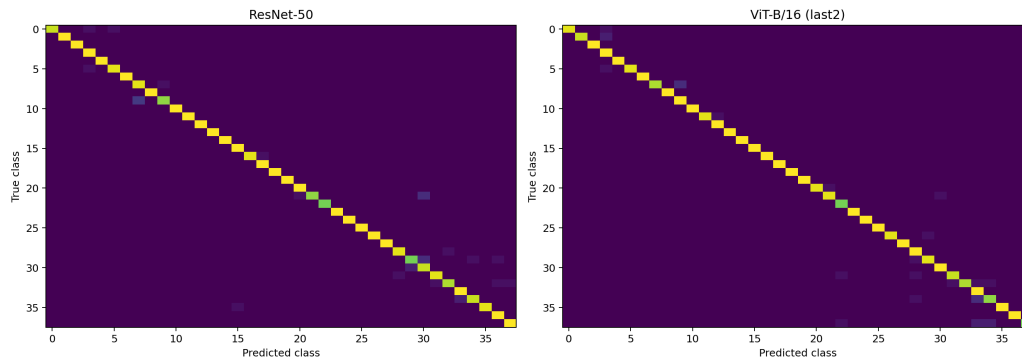


Figure 2: Final test confusion matrices for the selected ViT-B/16 and ResNet-50.

5.2 Per-Class Disagreement and Representative Errors

The per-class analysis identified several classes where the ViT had higher F1 than ResNet. The largest reported ViT advantages were:

- Tomato Late blight: +0.133 F1,
- Tomato Early blight: +0.114,
- Corn Northern Leaf Blight: +0.052,
- Potato Late blight: +0.049.

The notebook also reported smaller positive differences for Tomato mosaic virus, Corn Cercospora/Gray leaf spot, Apple scab, and Tomato Yellow Leaf Curl Virus. These are class-specific differences and should not be interpreted as a general ViT advantage because the overall paired test was inconclusive.

The representative disagreement table provides concrete failure cases. Among the examples where ResNet was correct and ViT was wrong, Apple scab was predicted by ViT as Apple healthy, Potato Early blight was predicted as Potato Late blight, and Tomato Target Spot was predicted as Tomato Spider mites. The exported representative table also contained an example in the opposite direction: an Apple scab image was classified correctly by ViT while ResNet predicted Apple healthy. These examples show that the two architectures can make different decisions even when their aggregate test scores are almost identical.

5.3 Interpretability: ViT Attention versus ResNet Grad-CAM

The notebook directly implemented the interpretability comparison requested by the handbook. For ViT, final-block attention was captured and the CLS-to-patch weights were averaged across heads. For ResNet, Grad-CAM was computed from the final residual stage for the model's predicted class [4]. Both maps were resized to the input image resolution and displayed on the same test examples.

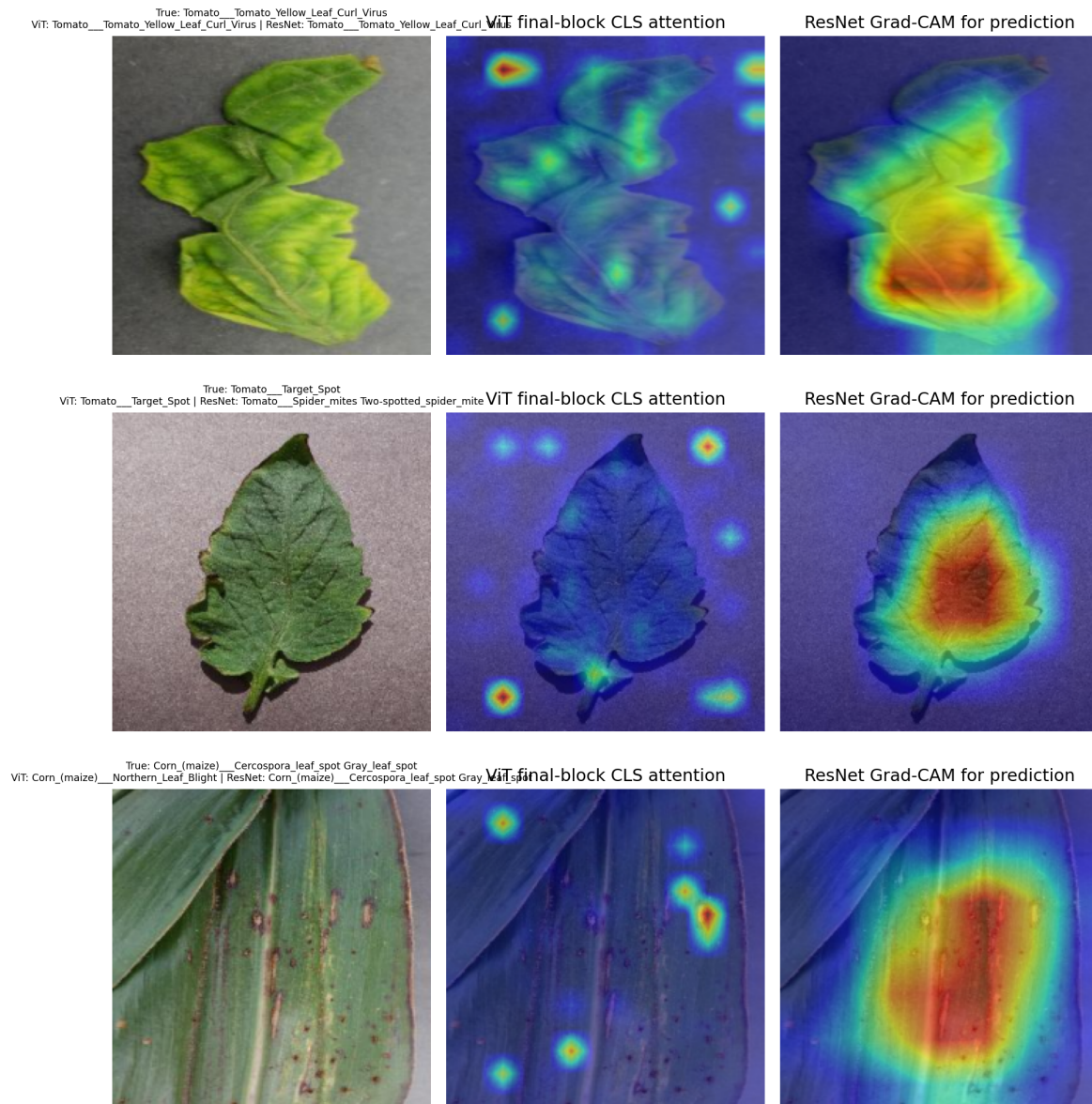


Figure 3: The same test images shown with selected-ViT final-block CLS-to-patch attention and ResNet-50 Grad-CAM.

The displayed examples show a clear qualitative difference in spatial organization. The ViT maps are sparse and patch-like, with several separated hotspots; in some examples they also assign visible attention outside the leaf. The ResNet Grad-CAM maps are broader and more spatially contiguous, concentrating strongly on central leaf regions. For the Tomato Yellow Leaf Curl Virus example, both models predicted the correct class, but the ViT distributed attention over several separated locations while Grad-CAM concentrated on a broad lower-leaf region. In the Tomato Target Spot disagreement example, ViT was correct and ResNet predicted Spider mites; the two explanations again emphasized different spatial patterns. In the corn Cercospora/Gray leaf spot example, ResNet was correct while ViT predicted Northern Leaf Blight, and the attention/activation patterns diverged. These observations are qualitative: CLS attention and Grad-CAM are different explanatory quantities and neither is treated as causal evidence.

6 Stretch Bonus: Data-Efficiency Sweep

The ViT-1 handbook card specifies an accuracy-versus-training-set-size sweep for ViT and CNN as a stretch experiment. The notebook implemented this with lightweight head-only variants. For each architecture, the pretrained backbone was frozen and only the classifier head was trained for one epoch on the same stratified training fractions.

Table 3: Stretch data-efficiency experiment.

Model	Fraction	Training images	Validation accuracy
ResNet-50 head-only	0.25	1,057	0.6645
ViT-B/16 head-only	0.25	1,057	0.8079
ResNet-50 head-only	0.50	2,114	0.8024
ViT-B/16 head-only	0.50	2,114	0.9084
ResNet-50 head-only	1.00	4,228	0.8510
ViT-B/16 head-only	1.00	4,228	0.9305

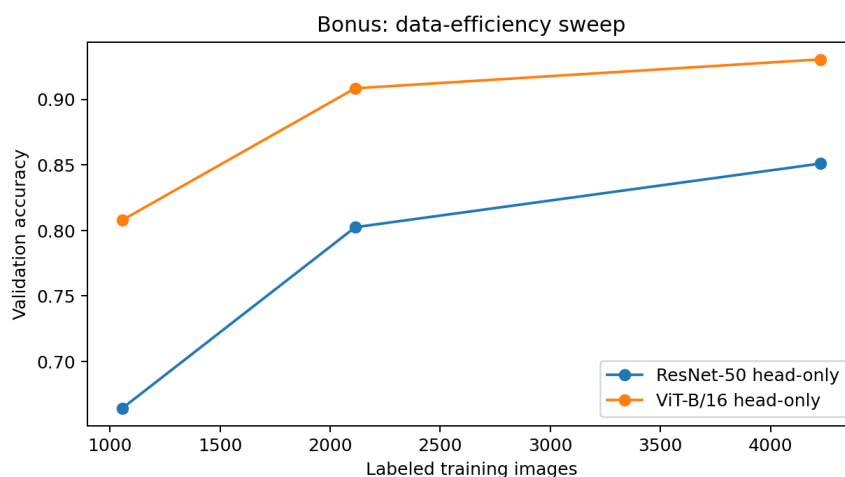


Figure 4: Validation accuracy versus labeled training size for the head-only ViT-B/16 and ResNet-50 stretch experiment.

Under this deliberately lightweight head-only protocol, ViT achieved higher validation accuracy at every tested fraction. This result is supplementary and should not be conflated with the main experiment, where ResNet was fully fine-tuned and the selected ViT used its final two blocks.

7 Discussion and Observation

7.1 Question 1: Where Should Attention Be Added to a CNN?

The handbook asks whether attention should be introduced early or late in a CNN and what change in learned features should be expected. The notebook answered this experimentally by

placing self-attention *late*, after a frozen pretrained ResNet-50 convolutional backbone.

Both supplementary classifiers started from the same frozen backbone and used the same training and validation data. The CNN control projected the final 7×7 feature map to 256 channels, normalized the resulting 49 spatial tokens, mean-pooled them, and classified them. The hybrid used the same projection but inserted four-head multi-head self-attention over the 49 tokens, with a residual connection and post-attention normalization, before pooling and classification [5]. Only the lightweight heads were trainable.

Table 4: Controlled late-attention experiment on the validation split.

Model	Accuracy	Macro-F1	Trainable params	Seconds
CNN control	0.9536	0.9535	534,822	36.8746
CNN + late attention	0.8996	0.8931	798,502	37.7256

Late attention reduced validation Macro-F1 by 0.0604. Therefore, the experiment provides no evidence of a classification-performance benefit from the added attention block under this setup. However, the learned attention had a mean spatial interaction distance of 3.726 cells on the 7×7 feature map, showing that the module combined information across spatially separated regions rather than behaving like another purely local operator.

The rationale for late placement was to preserve the CNN’s early local and translation-equivariant extraction of edges, textures, and lesion-like patterns, while allowing higher-level spatial features to interact globally. The result supports the expected representational change toward longer-range interaction but is a **negative performance result**: for this PlantVillage subset and training budget, the plain CNN control was more effective.

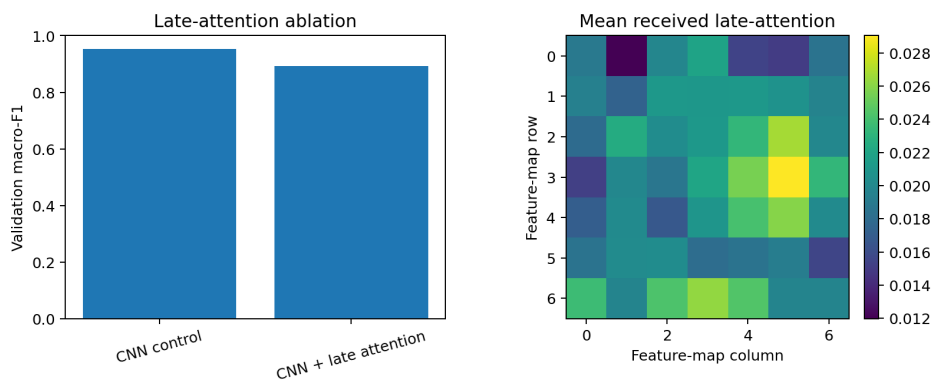


Figure 5: Late-attention ablation and the learned received-attention pattern on the final 7×7 feature map.

7.2 Question 2: What Do ViT Attention and CNN Grad-CAM Focus On?

The side-by-side experiment in Figure 3 directly addresses the second handbook question. The ViT visualization represents final-block CLS attention to image patches, whereas Grad-CAM

is class-specific and highlights regions that support the ResNet prediction. Consequently, the maps should not be interpreted as measuring exactly the same quantity.

In the displayed notebook examples, ResNet Grad-CAM generally forms broad contiguous responses over leaf tissue, while ViT attention is distributed across discrete patches and can include separated regions. This matches the architectural intuition qualitatively: convolution preserves a strong local spatial bias, whereas self-attention can relate non-adjacent patches. The disagreement examples are especially informative because the two models not only visualize different regions but also predict different classes. Nevertheless, these maps remain explanatory probes rather than causal demonstrations of the models' reasoning.

8 Limitations and Next Steps

PlantVillage contains relatively clean imagery compared with field conditions. The notebook also used a capped per-class stratified subset for Kaggle practicality, so the original dataset prevalence was not preserved. The compact two-epoch main budget measures behavior under a controlled regime rather than each architecture's fully optimized performance.

Although the main experiment matched data split, preprocessing, image size, batch size, epoch budget, optimizer family, and validation-selection rule, it did not equalize FLOPs or trainable parameter counts. The layer-unfreezing conclusion is therefore specific to this experimental setup. Likewise, the final ViT point estimate was only slightly higher than ResNet, and paired testing did not establish a reliable test-set difference.

The attention visualization is final-block CLS-to-patch attention averaged across heads, while Grad-CAM is class-specific. Their comparison is qualitative, and neither should be treated as causal evidence. The bonus data-efficiency study used head-only models for one epoch and therefore answers a narrower question than the main fine-tuning experiment.

Useful next steps are to repeat the unfreezing study over multiple random seeds, extend the epoch budget, evaluate stronger augmentations, test uncapped class distributions, and assess robustness on field-like or corrupted imagery. A parameter- or compute-matched comparison would also separate architectural effects from differences in trainable capacity more tightly.

References

- [1] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

- [2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.
- [3] David P. Hughes and Marcel Salathe. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv preprint arXiv:1511.08060*, 2015.
- [4] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 618–626, 2017.
- [5] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.